

Serpentine3D

Freeform NURBS surface modelling, open source at last.
Linux, Windows, macOS. MIT licensed.

自由曲面 NURBS 建模，終於有了開源選項。
Linux、Windows、macOS 全平台，MIT 授權。

v0.6.0

MIT

Linux

Windows

macOS

I design sets for film. I work on Linux. I was rebooting into Windows for one program.

我是影視場景設計師，平常在 Linux 上工作。
卻得為了一套軟體，一直重開機進 Windows。

So I built the program. It is named after serpentine, the stone Southern African sculptors carve, and after the curves NURBS is for.

於是我自己做了一套。名字來自南部非洲雕刻用的蛇紋石，也來自 NURBS 最擅長的流動曲線。

Open source has meshes. It has parametric CAD. Freeform NURBS surfaces, it never really had.

開源世界有網格建模，也有參數化 CAD。
唯獨自由曲面 NURBS，一直沒有真正的選擇。

MESHES / 網格 ✓

Blender

多邊形與細分曲面

PARAMETRIC CAD / 參數化 CAD ✓

FreeCAD

工程零件與約束

FREEFORM NURBS / 自由曲面 ✕

Rhino · Plasticity

只有商業軟體，沒有開源選項



Real BREP / NURBS geometry. Not meshes.

真正的 BREP / NURBS 幾何，建構於 OpenCASCADE 核心之上，不是三角網格。

WHAT IT ACTUALLY DOES / 它實際能做什麼

A full NURBS toolset, not a demo.

完整的 NURBS 工具鏈，不是展示品。

- Lofts, sweeps, revolves, booleans, shells, variable-radius fillets

放樣、掃掠、旋轉、布林、薄殼、變半徑倒角

- Object snaps, layers, gumball, sub-object select

物件鎖點、圖層、Gumball、次物件選取

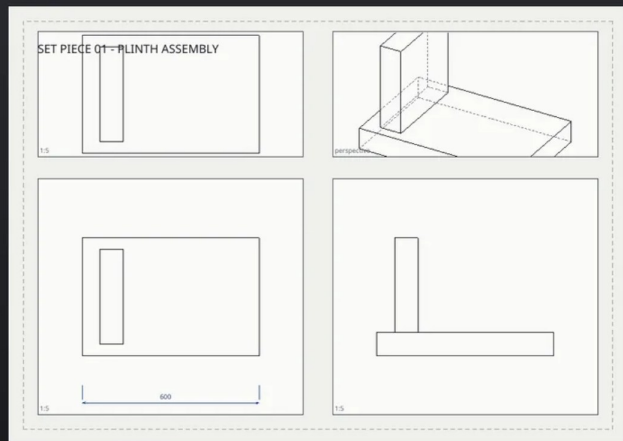
- Rhino-style command line, imports your Rhino aliases

Rhino 風格指令列，可匯入你原本的別名與快捷鍵

- STEP, OBJ, DXF, GLB, USD, SVG and .3dm, in and out

STEP、OBJ、DXF、GLB、USD、SVG、.3dm 匯入匯出

IT DOES NOT STOP AT THE MODEL / 不只是建模



Drawings and camera, in the same file.

圖說與鏡頭，都在同一個檔案裡。

- Paper layouts, detail views, hidden line removal
圖紙配置、細部視圖、隱藏線消除
- Live sections, associative dimensions, vector PDF
即時剖面、關聯式標註、向量 PDF
- Cinema lens presets: Super35, Alexa LF, 65mm
電影鏡頭預設與構圖框：Super35、Alexa LF、65mm

AI NATIVE, NOT AI ADDED / 從一開始就為 AI 設計

A built-in MCP server, from day one.

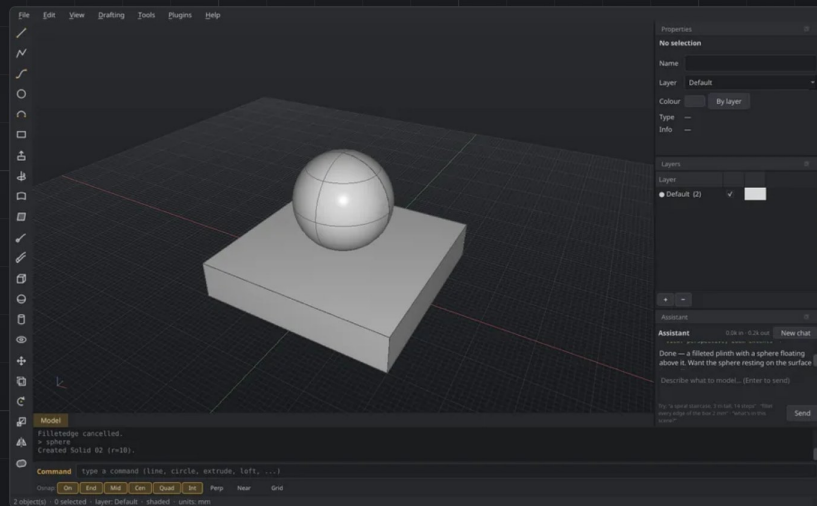
內建 MCP 伺服器，從第一天就在。

Any coding agent can see your viewport, build geometry and run commands beside you. Claude, Cursor, Codex, whatever you already use.

任何 AI 代理都能看見你的視埠、建立幾何、與你並肩下指令。

First open source CAD tool to ship one built in.

第一套內建 MCP 的開源 CAD。



While you model, the session writes itself down.

你在建模的同時，這段過程正把自己寫成一份日誌。

```
~/.local/share/serpentine3d/journals/20260809-024129.jsonl  
{ "ev": "session", "ver": 1, "app": "0.6.0" }  
{ "ev": "cmd", "name": "circle", "sel": [] }  
{ "ev": "val", "v": { "p": [0.0, 0.0, 0.0] } }  
{ "ev": "val", "v": { "p": [20.0, 0.0, 0.0] } }  
{ "ev": "fin", "ok": true, "made": [ "24f4f0cc" ] }  
{ "ev": "cmd", "name": "loft", "sel": [ "24f4f0cc", "710d5efb" ] }  
{ "ev": "fp", "objects": [ /* geometry fingerprint */ ] }
```

So a model becomes **a recipe**.

模型不再只是一坨幾何，而是一份可以重跑的「食譜」。

```
$ serp3d replay 20260809-024129.jsonl --check
replayed 14 events -> 3 object(s)
2 fingerprint(s) match: the replay is faithful

$ serp3d replay 20260809-024129.jsonl --video build.mp4 --aspect 9:16
```

Headless in CI, diffable, version controlled, and it renders its own timelapse. No screen recorder.

可無介面執行、可比對差異、可版本控制，還能自己算出縮時影片，不需要螢幕錄影。

COME AND BUILD IT / 一起來做

MIT licensed. Three platforms. One person so far.

MIT 授權，三大平台皆有安裝檔。目前只有我一個人在做。

0.6.0

shipped today
今天發布

2026

tests passing
個測試全數通過

3

installers
個平台安裝檔

35k

lines of Python
行 Python

github.com/chisomobanzi/Serpentine3D

Thank you · 謝謝大家



SCAN / 掃描